

Pressurisation

1. a) Расскажите о предназначении Overboard Exhaust Valve (OEV).

b.) Какую конфигурацию необходимо создать для улучшения вентиляции при удалении дыма?

Ответ:

a.) The B737 **Overboard Exhaust Valve (OEV)** is part of the aircraft's equipment cooling system. It is located in the aft center section of the Equipment and Electrics (E&E) compartment and regulates the exhaust flow of conditioned air used to cool the avionics and flight deck instruments.

b.) Overboard exhaust valve is driven open if either pack switch is in high and the right recirculation fan is off. This allows for increased ventilation in the smoke removal configuration.

2. a.) Расскажите о работе системы наддува в Auto Mode Operation на эшелоне?

Практический вопрос:

b.) Самолет летит на эшелоне 370 (в окошке FLT ALT 370), смена эшелона на 380. Что произойдет с высотой в кабине при занятии эшелона 380 и переустановки FLT ALT 380?

Ответ:

a.) The cruise mode is activated when the airplane climbs to within 0.25 psi of the selected FLT ALT. During cruise the controller maintains the lowest possible cabin altitude based on the differential pressure limits indicated in the table below.

b.) 28,000 feet to 37,000 feet>>>7.80 psid

Above 37,000 feet>>>>8.35 psid

37000

За бортом (на FL370): Давление воздуха в стандартной атмосфере (ISA) составляет всего **3.14 psi** (около 162 мм рт. ст.)

В кабине: $3.14 + 7.80 = 10.94$ psi

Абсолютное давление по барометрической таблице соответствует высоте 8000ft над уровнем моря (2438м)

На эшелоне 380:

Атмосферное давление 2.99

Перепад 8.35

$2.99 + 8.35 = 11.34$ >> соответствует 7000 ft (2130м)

Вывод:

Высота в кабине снизится, несмотря на набор высоты!!!

Electrical

1. Расскажите про селекторы на AC and DC metering panel (нормальные значения при каждом положении селектора)?

AC meters:

- all position of AC meter selector should read 400 cycles +/- 10 and 115v +/- 5 if everything is normal.
- INV position lets you monitor the output of the static inverter
- STBY PWR shows frequency and voltage of AC standby bus.

DC meters:

- all meter selector position you are reading a TR output so you should see 28v. Normal range is 24-30v
- loads indicated only on BAT, TR1, TR2, TR3
- BAT/AUX BAT 28v (battery charger is working), less than 28v (is not working)

Note: main BAT may be vary +20 to +40 for short period after APU start

- STBY PWR shows volts of DC Standby bus

2. Расскажите об индикации GEN OFF BUS и APU GEN OFF BUS?

GEN OFF BUS light:

- illuminated means IDG is not supplying power to related transfer bus

- illuminated when GCB (generator control breaker) is open

APU GEN OFF BUS light:

- illuminated means APU is up to speed (95%), frequency is above 380 Hz and ready for load, however, the APU is not power source in use
- APB (aux power breaker) is open

Note: after start if light fails to illuminate, the APU may have failed to reach starter cutout speed. A hung start may occur.

Hydraulic

1. Расскажите о работе Thrust Reverser с точки зрения работы гидросистем (Normal operation and hydraulic failure)?

Hydraulic pressure for the operation of engine No. 1 and engine No. 2 thrust reversers comes from hydraulic systems A and B, respectively. If hydraulic system A and/or B fails, alternate operation for the affected thrust reverser is available through the standby hydraulic system. When the standby system is used, the affected thrust reverser deploys and retracts at a slower rate and some thrust asymmetry can be anticipated.

2. Расскажите о работе Parking Brake с точки зрения работы гидросистем (какую систему использует, будет ли сохраняться давление при выключенных системах А и В, как проконтролировать давление, признаки отказа)?

The parking brake can be set with either A or B hydraulic systems pressurized. If A and B hydraulic systems are not pressurized, parking brake pressure is maintained by the brake accumulator. Accumulator pressure is shown on the HYD BRAKE PRESS indicator.

A fault in the parking brake system may cause the ANTISKID INOP light to illuminate.

Fire protection

1. Расскажите о работе Engine overheat and Fire detection (loop, sensors, location, conditions(temperature)).

Each engine has eight detectors. The detectors monitor four sections of the engine. In each section, two detectors attach to a support tube and make an assembly. An assembly has one detector from loop A and one from loop B.

These are the locations of the engine fire detectors:

- Two on the upper fan case section
- Two on the lower fan case section
- Two on the left core section
- Two on the right core section.

Upper Fan Case: 345F (174C) / 580F (304C)

Lower Fan Case: 345F (174C) / 580F (304C)

Left Core Section: 650F (343C) / 850F (454C)

Right Core section: 650F (343C) / 850F (454C)

2. Расскажите о работе APU Fire Detection (loop, sensors, location, conditions(temperature))

The APU has three detectors. The detectors monitor three sections of the APU. In each section, one detector attaches to a support tube with clamps.

There is one loop for the APU fire detection.

These are the locations of the APU fire detectors:

- Upper APU compartment
- APU compartment door
- Tailpipe.

Upper APU Compt 450F (232C)

Lower APU Compt 450F (232C)

Tailpipe 275F (135C)

3. Расскажите о работе Cargo Fire Warning and Smoke Detection (loop, sensors, location, conditions(temperature))

The lower cargo compartments have many cargo smoke detectors and one cargo electronic unit in each compartment.

The forward lower cargo compartment has four smoke detectors. The aft lower cargo compartment has six smoke detectors. The detectors are in the ceiling of the compartments.

The smoke detectors use photoelectric cells to detect smoke. The smoke detector gives an alarm signal if it senses smoke or if it senses air temperature more than 230F (110C).

AUTOFLIGHT

Вопросы:

a.) Особенности работы ROLL MODES AFDS на взлете. Hsel TO option, условия армирования LNAV перед взлетом. Что покажет FD's по roll до 50', до 400', после 400' в каждом режиме (ограничения по крену)?

b.) Практический пример: Команда диспетчера: «На курсе взлета набирайте эшелон/высоту.../ After departure maintain RW heading, climb...».

Какой режим работы выберете на MCP? Необходима ли поправка на ветер?

Ответы:

a.) 73313,73269,73315,73316,73317,73318,73319,73321 (SB Changes 73314)

During takeoff, prior to 60 KIAS:

- the roll command is wings level

73311,73312 (SB Changes 73314) During takeoff, prior to 60 KIAS:

- the roll command is HDG SEL

At LIFT-OFF:

73313,73269,73315,73316,73317,73318,73319,73321 (SB Changes 73314) • the roll command maintains wings level.

73311,73312 (SB Changes 73314) • the roll command maintains HDG SEL. Bank angle is limited to 8 degrees below 400 feet, and 10–30 degrees selectable above 400 feet AGL.

“FCOM 4.20.11 Automatic Flight Takeoff and Landing”

LNAV arming criteria on the ground:

- origin runway in flight plan
- active route entered in FMC
- track of first leg within 5 degrees of runway heading
- both flight director switches have been switched on.
- LNAV selected prior to TO/GA.
- LNAV guidance becomes active at 50 feet AGL

73313,73269,73315,73316,73317,73318,73319,73321 (SB Changes VP-73313) • bank angle is limited to 8 degrees below 200 feet and 30 degrees above 200 feet AGL.

73311,73312,73314,73315,73319 (SB Changes 73313) • bank angle is limited to 15 degrees below 200 feet and 30 degrees above 200 feet AGL.

“FCOM 4.10.15 Lateral Navigation”

b.) RUNWAY HEADING– The magnetic direction that corresponds with the runway centerline extended, not the painted runway number. When cleared to “fly or maintain runway heading,” pilots are expected to fly or maintain the heading that corresponds with the extended centerline of the departure runway. Drift correction must not be applied; e.g., Runway 4, actual magnetic heading of the runway centerline 044, fly 044.

Doc.: “FAA Order JO 7110.65BB”